

DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

WW-BR-603

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	603	1MF	OD400-WW-BR-HP1-N-7022-01	1 OF 1	1	CJ-M15W14-PR-A20-000503,504	1A	

ITEM NO.	W14-T-4726
SERVICE	BRINE 2'nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 4.1 x 5.775 x 6.6 (EH 5.0)
CAPA.	m3 156.3 (EH 118.4)
HRT	min 34.1
MAT'L	P.C + FRP

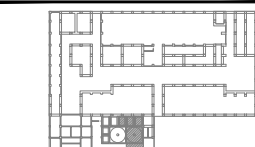
ITEM NO.	W14-T-4727
SERVICE	BRINE 2'nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 4.2 x 5.775 x 6.6 (EH 5.0)
CAPA.	m3 160.1 (EH 121.3)
HRT	min 34.9
MAT'L	P.C + FRP

ITEM NO.	W14-T-4728
SERVICE	BRINE 2'nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 4.7 x 5.775 x 6.6 (EH 5.0)
CAPA.	m3 179.1 (EH 135.7)
HRT	min 39.1
MAT'L	P.C + FRP

ITEM NO.	W14-T-4729
SERVICE	BRINE 2'nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 9.225 x 9.225 x 6.6 (EH 5.0)
CAPA.	m3 561.7 (EH 425.5)
HRT	hr 2.0
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4729
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 5,000m3/d)
MAT'L	ABS+SS304

대외비



KEY PLAN (1MF)

NOTE

1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
5. 온천 시 CLOSE 되는 배관은 LOTO 설치.
6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
7. AGITATOR MAT'L 표기 : SHAFT/BLADE
8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 반영 할 것.
(SLAB으로부터 100mm 떨어질 것)
9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
10. 위에 것수등의 상세사상은 추후 VENDOR에서 결정한다.
11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
12. FLUSHING 용 WATER / AIR 배관은 45도 서언으로 연결할 것.
13. SLUDGE REMOVAL PUMP 또는 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향상.
14. DELETED.
15. 추후 VENDOR에서 결정한다.
16. DELETED.
17. DELETED.
18. DELETED.
19. DELETED.
20. SLUDGE PUMP는 TIMER에 의해 온전됨.

330

603

376

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31

REV.	실명	날짜
1	이인환	
2	한진희	
3	박동명	

SK 에코플랜트

PROJECT TITLE

M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
BRINE 2'nd COAGULATION & SETTLING TANK

속척
SCALE
NONE

날짜
DATE
2025. 08. 14

PROJECT NO. 22330D
도면번호
SHEET NO. CJ-M15W14-PR-A20-000503
REV. 1A

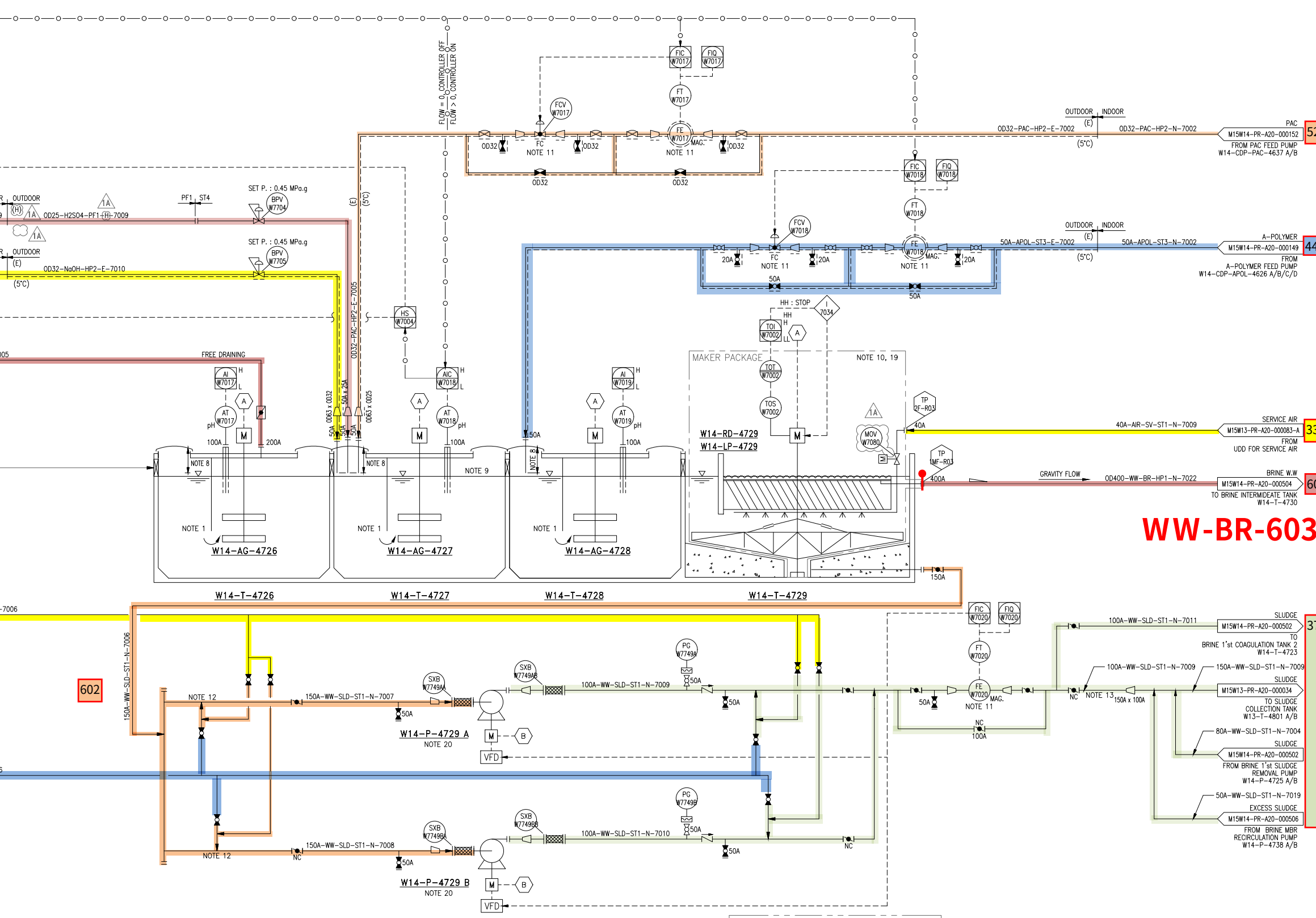
ITEM NO.	W14-AG-4726
SERVICE	BRINE 2'nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (37)
MAT'L	STS316L/STS316L
POWER	kw (1)

ITEM NO.	W14-AG-4727
SERVICE	BRINE 2'nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (30)
MAT'L	STS316L/STS316L
POWER	kw (7.5)

ITEM NO.	W14-AG-4728
SERVICE	BRINE 2'nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (3.7)

ITEM NO.	W14-RD-4729
SERVICE	BRINE 2'nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/d (0.079)
MAT'L	STS304/STS304
POWER	kw (1.5)

ITEM NO.	W14-P-4729 A/B
SERVICE	BRINE 2'nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVF), VORTEX
CAPA.	m3/hr.ea x mH 40 x 30 (0.29MPa)
MAT'L	SSC13/SSC19
POWER	kw (30)



WW-BR-603

C:\Users\WTOP\Desktop\WP&ID251013\W15X Ph-4 for 인허가\CAD FILE_0910\WCJ-M15W14-PR-A20-000503 R1 P&ID FOR BRINE

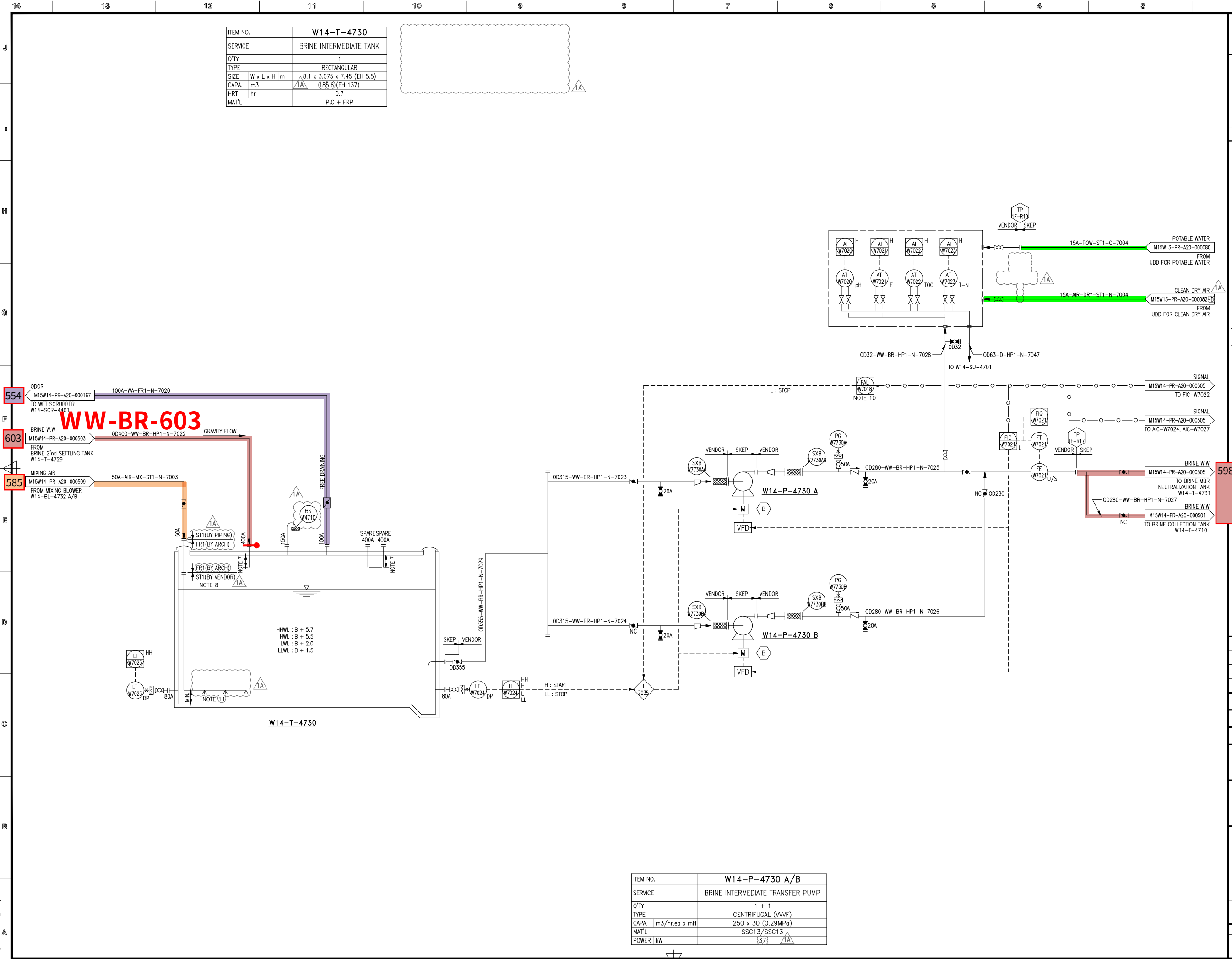
ITEM NO.	W14-T-4730
SERVICE	BRINE INTERMEDIATE TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 8.1 x 3.075 x 7.45 (EH 5.5)
CAPA.	m3 /A (185.6) (EH 137)
HRT	hr 0.7
MAT'L	P.C + FRP



대외비

KEY PLAN (1F)

- NOTE
1. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 별반 설치할 것
 2. 환수 인입지점은 펌프 흡입측에서 최대한 이격하여 설치
 3. 2m 이상 높은 곳에 설치된 별반은 CHAIN VALVE 또는 PLATFORM 구성할 것
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 콘센트 CLOSE 되는 배관에는 LOTO 설치
 6. CENTRIFUGAL PUMP
MAT'L 표기 : CASING / IMPELLER
 7. TANK 내 유입, 유출, 유입이 가능하고, 비산되지 않도록 최소 길이로 반영할 것 (SLAB 로 부터 100mm 떨어질 것)
 8. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 9. INTERLOCK NO. I-7035
(W14-P-4730 A/B AUTO STOP)
: REMOTE STATUS에 W14-T-4730 LEVEL (LI-W7024) LOW LOW STOP 또는 W14-P-4730 A/B DISCHARGE FLOW (FAL-W7021) LOW STOP.
 10. BRINE INTERMEDIATE TRANSFER PUMP RUN 후 30초 후 부터 동작한다.
 11. (DIFFUSER HOLE 및 배관배치의 관련된 상세 설계는 VENDOR가 수행한다.)



ITEM NO.	W14-P-4730 A/B
SERVICE	BRINE INTERMEDIATE TRANSFER PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVVF)
CAPA.	m3/hr.ea x mH 250 x 30 (0.29MPa)
MAT'L	SSC13/SSC13
POWER kw	{37} /A

598

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31
수정 REV.	설명 DESCRIPTION	날짜 DATE

제 도	이 인 학
DRAWN BY	
검 토	한 진 희
CHECKED BY	
승 인	박 동 명
APPROVED BY	
제 작 사	

SK 에코플랜트

공 사 명
PROJECT TITLE
M15X Ph4 WWT Project

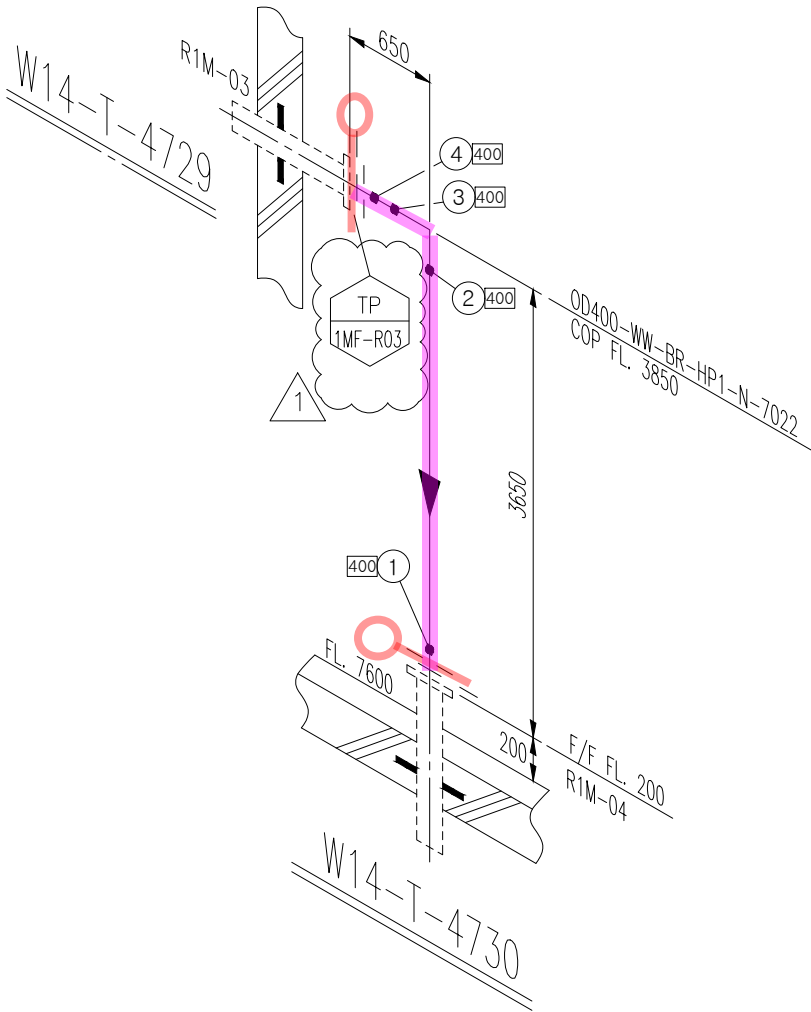
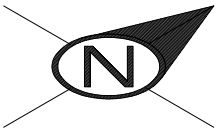
도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
BRINE INTERMEDIATE TANK &
TRANSFER PUMP

속 척
SCALE
NONE

날 짜
DATE
2025. 08. 14

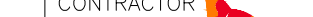
PROJECT NO.	22330D	REV.	
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000504	1A	

A1(594x841mm 1:2mm)



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	6.4	60	UNIT	R1MF								M15X Ph4 Project		
			OPERATING	4	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-010018-C		1	25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	WW-BR	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000503,000504		0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	OD400-WW-BR-HP1-N-7022	SH'T NO. 01 / 01	REV. 1
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				